

Joint Modulation and Deep Learning-Based BER Optimization in MIMO Molecular Communication Systems Under ISI and ILI

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Abstract—Molecular Communication (MC) is a bio-inspired approach towards nano-scale communication networks wherein information is conveyed utilizing diffusing molecules. The efficiency of Multiple-Input Multiple-Output (MIMO) molecular communication can be adversely impacted by Inter-Symbol Interference (ISI) and Inter-Link Interference (ILI). In this paper, an existing joint modulation scheme has been studied and incorporated into a Deep Neural Network (DNN) based optimization scheme for a 2×2 diffusion-based MIMO MC network. The joint modulation strategy uses multi-molecule type diversity in terms of time and space to overcome interference problems in highly correlated transmission slots. Analytical expressions for molecule concentration, received signal statistics, ISI, ILI and Bit Error Rate (BER) have been derived through Fick's second law, Gaussian approximation, and Maximum A Posteriori (MAP) based threshold detection. Additionally, a DNN model has been employed to optimize the number of transmitted molecules in order to minimize BER. Simulation results show that the combination of DNN optimization and joint modulation provides significant BER improvement while efficiently eliminating ISI and ILI in the 2×2 MIMO molecular communication system.

Index Terms—Molecular communication, MIMO, joint modulation, ISI, ILI, deep neural network, BER optimization.

I. INTRODUCTION

Due to the exponential growth of nanotechnology, nanoscale devices have been developed that can sense, compute, and communicate within biological systems. The use of conventional electromagnetic communications would be ineffective within these kinds of environments, because of strong signal attenuations, size constraints of the antenna, and biocompatibility issues while operating within fluidic and biological environments [1]. Molecular Communication (MC) presents itself as an effective method for transmitting information with the use of molecules of different types, including proteins, ions, and even Deoxyribonucleic acid (DNA) strands [2]. Diffusion-based MC uses molecules emitted from the transmitter and transferred to the receiver by Brownian movement. As the process of molecule transfer does not require any additional source of power, it is considered very energy-efficient and biocompatible.

Multiple-Input Multiple-Output (MIMO) molecular communication enhances reliability and throughput by using more than one transmitter-receiver combination at a time [1]. The

standard 2×2 MIMO system consists of two transmitters and two receivers, which provide several diffusion paths so that data can be sent in parallel streams [3]. However, due to the randomness of molecule diffusion, there is a high probability of interference in MIMO systems. Molecules released in previous time slots may arrive in the current slot, resulting in Inter-Symbol Interference (ISI) [4]. Additionally, molecules released by one transmitter may be absorbed by a neighboring unintended receiver, producing Inter-Link Interference (ILI) [5].

A number of methods have been suggested to mitigate ISI and interference in MC [6]. Concentration Shift Keying (CSK) utilizes concentration levels of molecules to encode symbols; however, while CSK is simple to implement, it does not deal well with ISI because residual molecules interfere with subsequent slots [7]. Molecule Shift Keying (MoSK) employs different molecule types to encode distinct symbols, offering improved interference resilience [7]. Other modulation techniques that handle ISI avoid using the same molecule type in adjacent slots [8]. Spatial and index modulation techniques have also been developed for MIMO molecular communication to increase reliability and spectral efficiency [8], [9].

In recent times, there has been an increased interest in modulation schemes that are interference-aware, as well as intelligence-based optimization methods. The existing joint modulation architecture for 2×2 MIMO MC systems makes use of several molecule types and correlated transmission slots to minimize the effects of ISI and ILI [2]. Such schemes exploit both spatial and temporal diversity so that the same molecules do not interfere with each other in neighboring links and time slots [10]. On the other hand, DNN optimization algorithms for MC have attracted considerable interest owing to their ability to model channel nonlinearities [11]. Optimization algorithms using neural networks have outperformed traditional analytical approaches both in terms of BER performance and computation time [12].

While joint modulation schemes and DNN-based optimization have individually been studied, their joint evaluation in the same 2×2 MIMO diffusion-based MC framework has not been widely addressed [2], [11]. Some previous studies focused on minimizing interference using modulation alone [13],

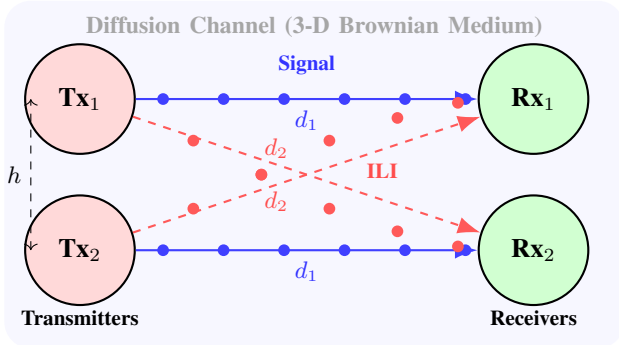


Fig. 1: System architecture of the proposed 2×2 MIMO diffusion-based MC system. Solid blue arrows indicate desired signal links (d_1); dashed red arrows represent ILI paths (d_2).

while others considered machine learning based parameter optimization separately [14]. A unified ISI/ILI BER analysis framework combined with DNN-based transmission parameter optimization has not yet received due attention. Prompted by these observations, this study examines the joint modulation framework of [2] and combines it with a DNN optimizer for BER enhancement in a 2×2 diffusion-based MIMO MC system.

The main contributions of this work are as follows:

- The performance of an existing joint modulation framework for a 2×2 MIMO diffusion-based MC system is analyzed, focusing on its capability to mitigate both ISI and ILI through spatial and temporal molecule diversity.
- Analytical expressions for molecule concentration, ISI, ILI, and BER are derived using Fick's law, Gaussian approximation, and MAP-based threshold detection under the considered joint modulation scheme.
- A DNN-based optimization framework is developed that learns the relationship between channel conditions and optimal molecule transmission parameters for BER minimization in the 2×2 MIMO system.
- The combined impact of joint modulation and DNN-based optimization is evaluated through Monte Carlo simulations, demonstrating significant BER improvement.

II. SYSTEM MODEL

We consider a 2×2 MIMO diffusion-based MC system operating in an unbounded three-dimensional fluid medium, as illustrated in Fig. 1. The system consists of two point transmitters, Tx_1 and Tx_2 , and two corresponding spherical receivers, Rx_1 and Rx_2 [1]. The transmitters are separated by a distance h along the spatial axis, while each receiver is positioned at distance d_1 from its intended transmitter and at distance d_2 from the neighboring transmitter, thereby creating both direct and cross diffusion links [5]. Information is transmitted using molecules released instantaneously at the beginning of each symbol duration t_s [2].

The propagation of molecules follows isotropic Brownian diffusion characterized by diffusion coefficient D ($\mu\text{m}^2/\text{s}$)

[15]. All emitted molecules are assumed to be independent and chemically identical for a given molecule type. The receivers operate as passive spherical observers that count the molecules arriving within a symbol interval [15]. Perfect synchronization among all nodes is assumed, and the channel characteristics remain constant during a modulation cycle [3].

The probability that a molecule emitted from a transmitter at distance d reaches the corresponding spherical receiver of radius R within time t_s is expressed as [2], [4]

$$p_r(d, t_s) = \frac{R}{R + d} \operatorname{erfc}\left(\frac{d}{\sqrt{4Dt_s}}\right), \quad (1)$$

where $\operatorname{erfc}(\cdot)$ is the complementary error function. When N molecules are transmitted during a symbol interval, the count N_r of molecules observed at the receiver follows a binomial distribution. For sufficiently large N , this is approximated by the Gaussian distribution

$$N_r \sim \mathcal{N}(Np_r, Np_r(1 - p_r)), \quad (2)$$

where N_r is the number of received molecules, with mean Np_r and variance $Np_r(1 - p_r)$. Due to the stochastic nature of diffusion, molecules transmitted in previous slots may arrive in the current slot, causing ISI [6], while molecules from neighboring transmitters may reach unintended receivers, producing ILI [5]. The total received signal at receiver Rx_j during the k^{th} time slot is therefore

$$y_{j,m}[k] = N_{\text{desired}} + N_{\text{ISI}} + N_{\text{ILI}} + \eta, \quad (3)$$

where N_{desired} represents the desired received molecules, N_{ISI} and N_{ILI} denote the ISI and ILI components respectively, and η corresponds to receiver noise.

III. JOINT MODULATION TECHNIQUE

The joint modulation framework considered here is designed to mitigate both ISI and ILI in a 2×2 diffusion-based MIMO MC system [2]. The fundamental principle is that interference mainly occurs when identical molecule types appear in temporally adjacent slots or across neighboring transmitter–receiver links simultaneously [8]. To address this, the modulation strategy utilizes three distinct molecule types—A, B, and C—along with an idle symbol “–” representing no emission [2]. The transmission operates in correlated modulation cycles of multiple consecutive time slots [10]. Spatial and temporal diversity are jointly enforced so that identical molecule types are not reused across adjacent correlated slots or neighboring transmitter links [9]. For bit “1”, a transmitter selects an available molecule type from $\{A, B, C, -\}$ while ensuring the chosen type does not violate the no-repeat constraint across both temporal and spatial dimensions. For bit “0”, the idle symbol is transmitted. This coordinated mechanism significantly reduces the probability of molecule overlap responsible for ISI and ILI [2].

Fig. 2 illustrates the molecule allocation over a representative six-slot segment. The cycle ensures molecule types are rotated systematically across both transmitters, minimizing

	Correlated					
Tx	I	II	III	IV	V	VI
Bit₁	1	1	1	0	1	1
Mol₁	A	B	C	-	B	C
Bit₂	1	1	1	0	1	1
Mol₂	C	-	A	-	A	B

Time Slot →

Fig. 2: Proposed joint modulation for one six-slot segment (two cycles). Tx₁ (blue) and Tx₂ (red) rotate molecule types such that no type is repeated within a cycle on either link.

TABLE I: Joint Modulation Bit-to-Molecule Mapping (Sample Cycle)

Slot	Bit ₁	Mol (Tx ₁)	Bit ₂	Mol (Tx ₂)
I	1	A	1	C
II	1	B	1	-
III	1	C	1	A
IV	0	-	0	-

interference accumulation while maintaining reliable symbol distinguishability at the receivers [3].

Table I presents the bit-to-molecule mapping for one modulation cycle. The coordinated allocation across Tx₁ and Tx₂ prevents simultaneous reuse of identical molecule types, suppressing both intra-link and inter-link interference [2].

The modulation decision process at each transmitter is illustrated in Fig. 3. Before emission, the transmitter verifies whether the selected molecule type satisfies the temporal and spatial interference constraints [2]. If a conflict exists, the next available type is selected; otherwise, transmission proceeds normally.

Algorithm 1 summarizes the overall joint modulation procedure across correlated time slots.

IV. CHANNEL MODELING AND MATHEMATICAL ANALYSIS

The spatiotemporal concentration $C(\mathbf{r}, t)$ of information molecules in the fluid medium obeys Fick's second law [2], [15]:

$$\frac{\partial C(\mathbf{r}, t)}{\partial t} = D \nabla^2 C(\mathbf{r}, t). \quad (4)$$

Assuming a point source releasing M molecules at the origin at $t = 0$, the solution in free space is:

$$C(\mathbf{r}, t) = \frac{M}{(4\pi Dt)^{3/2}} \exp\left(-\frac{\|\mathbf{r}\|^2}{4Dt}\right). \quad (5)$$

Using the image-source method for a perfectly absorbing sphere of radius R at distance d , the probability that a single molecule is captured within slot duration t_s is expressed above in equation number (1). Where P_r converges to unity as $t_s \rightarrow \infty$ [4].

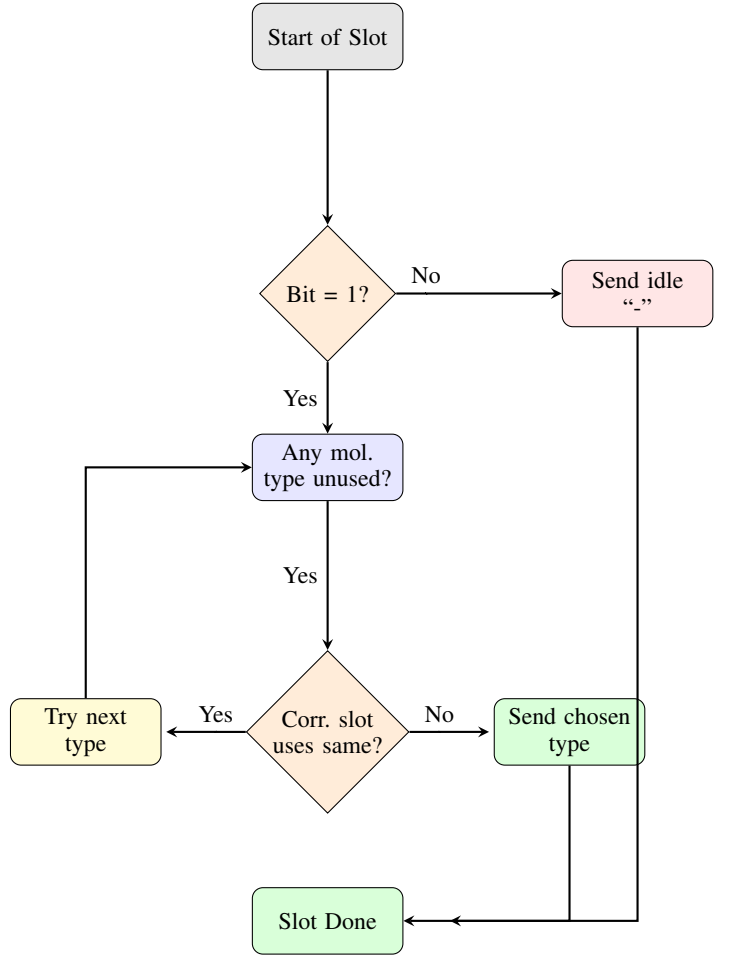


Fig. 3: Modulation decision flowchart for each transmitter at every time slot. Type selection enforces the no-repeat rule for both ISI and ILI avoidance.

The ISI contribution from molecules emitted i slots earlier, arriving at Rx₁ during the current slot, is the Gaussian variable [2], [6]:

$$N_{\text{ISI},i} \sim \frac{1}{2} \mathcal{N}\left(np(d_1, (i+1)t_s), np(d_1, (i+1)t_s)(1 - p(d_1, (i+1)t_s))\right) - \frac{1}{2} \mathcal{N}\left(np(d_1, it_s), np(d_1, it_s)(1 - p(d_1, it_s))\right), \quad (6)$$

and the total ISI summed over all previous slots is $N_{\text{ISI}} = \sum_{i=1}^{\infty} N_{\text{ISI},i}$. The total ILI from Tx₂ reaching Rx₁ (distance d_2) is [2], [5]:

$$N_{\text{ILI}} = N_{C,\text{ILI}} + \sum_{i=1}^{\infty} N_{P,\text{ILI},i}, \quad (7)$$

where $N_{C,\text{ILI}} \sim \mathcal{N}(np_r(d_2, t_s), np_r(d_2, t_s)(1 - p_r(d_2, t_s)))$.

The joint modulation rotation of three molecule types ensures that no more than 1/4 of slots on a given link are

Algorithm 1 Joint Modulation for 2×2 MIMO

Require: Bit sequences $\{b_1[k]\}$ for Tx₁, $\{b_2[k]\}$ for Tx₂

Ensure: Molecule type assignments $\{m_1[k], m_2[k]\}$

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1: Initialize used-type set  $\mathcal{U} \leftarrow \emptyset$  at start of each 4-slot cycle
2: for each slot  $k$  in order I, II, III, IV do
3:   for each transmitter  $i \in \{1, 2\}$  do
4:     if  $b_i[k] = 0$  then
5:        $m_i[k] \leftarrow \text{“—”}$  (idle)
6:     else
7:       Identify correlated slot  $k'$  (diagonal partner in cycle)
8:       for each type  $t \in \{A, B, C, -\}$  do
9:         if  $t \notin \mathcal{U}$  and  $t \neq m_{\bar{i}}[k']$  then
10:           $m_i[k] \leftarrow t$ ; break
11:        end if
12:      end for
13:    end if
14:     $\mathcal{U} \leftarrow \mathcal{U} \cup \{m_i[k]\}$ 
15:  end for
16: end for
17: return  $\{m_1[k], m_2[k]\}$ 
  
```

occupied by any single type [2]. Compared with a two-state CSK scheme, this reduces the ISI contribution by a factor of 1/4 per slot:

$$N_{\text{ISI, total}}^{(JM)} = \sum_{i=1}^{\infty} \frac{1}{4} [N(np_r(d, (i+1)t_s)) - N(np_r(d, it_s))]. \quad (8)$$

Similarly, ILI from adjacent links is reduced to [2]:

$$N_{\text{ILI, total}}^{(JM)} = \sum_{i=0}^{\infty} \frac{1}{4} [N(np_r(d_2, (i+1)t_s)) - N(np_r(d_2, it_s))]. \quad (9)$$

The desired received signal from Tx₁ at Rx₁ in the current slot follows [7]:

$$N_{\text{desired}} \sim \mathcal{N}(Nh_{11}[0], Nh_{11}[0](1 - h_{11}[0])), \quad (10)$$

where $h_{11}[0] = p_r(d_1, t_s)$ is the same-slot hitting probability for the desired link.

The receiver applies the MAP decision rule [11], [13]. Under the two hypotheses

$$H_0 : Z = N_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad (11)$$

$$H_1 : Z = N_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), \quad (12)$$

the likelihood-ratio condition leads to the quadratic threshold:

$$\tau = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}, \quad (13)$$

where

$$A = -\frac{1}{2} \left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_0^2} \right), \quad B = \frac{\mu_1}{\sigma_1^2} - \frac{\mu_0}{\sigma_0^2},$$

$$C = \ln \frac{\sigma_0}{\sigma_1} - \frac{1}{2} \left(\frac{\mu_1^2}{\sigma_1^2} - \frac{\mu_0^2}{\sigma_0^2} \right).$$

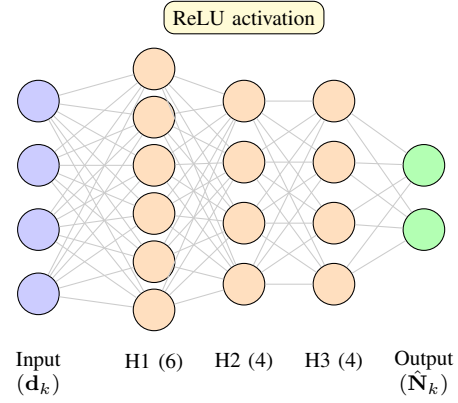


Fig. 4: DNN architecture for BER optimization. Input: pairwise transmitter–receiver distances $\mathbf{d}_k$; three hidden layers with ReLU activation; output: optimal molecule counts $\hat{\mathbf{N}}_k$ for each transmitter.

The means and variances under each hypothesis incorporate the contributions from (8)–(10):

$$\mu_1 = N_{\text{desired}}^{\mu} + N_{\text{ISI}}^{\mu, JM} + N_{\text{ILI}}^{\mu, JM}, \quad (14)$$

$$\sigma_1^2 = N_{\text{desired}}^{\sigma^2} + N_{\text{ISI}}^{\sigma^2, JM} + N_{\text{ILI}}^{\sigma^2, JM}. \quad (15)$$

Applying equal prior probabilities $P(H_0) = P(H_1) = 0.5$, the BER is [2], [4]:

$$\text{BER} = \frac{1}{2} Q\left(\frac{\mu_1 - \tau}{\sigma_1}\right) + \frac{1}{2} Q\left(\frac{\tau - \mu_0}{\sigma_0}\right), \quad (16)$$

where $Q(x) = \frac{1}{2} \text{erfc}(x/\sqrt{2})$. The threshold τ in (13) minimizes (16) under the MAP criterion.

V. DNN-BASED BER OPTIMIZATION

The DNN optimizer is introduced to address a fundamental limitation of the joint modulation scheme: even with interference-aware molecule-type selection, the number of molecules released by each transmitter remains a free parameter that directly controls the signal-to-interference ratio at each receiver. Fig. 4 illustrates the proposed DNN architecture, which takes the channel distance vector as input and outputs the optimal per-transmitter molecule counts. For a given channel distance vector $\mathbf{d}_k = [d_{11,k}, d_{12,k}, d_{21,k}, d_{22,k}]^T$, the goal is to find:

$$\mathbf{N}^* = \arg \min_{\mathbf{N}} P_e^{\text{avg}}(\mathbf{d}, \mathbf{N}), \quad (17)$$

subject to $\psi \leq N_i \leq \Psi$ for each transmitter i . The average BER P_e^{avg} is evaluated using (16) across all active links with ISI and ILI terms from Section IV. Because the feasible space grows exponentially with $\Psi - \psi$, exhaustive search is computationally prohibitive [16]. A DNN is therefore trained to learn the mapping $f_{\mathbf{w}, \mathbf{b}} : \mathbf{d} \mapsto \mathbf{N}^*$ [12].

The network comprises one input layer of size $M^2 = 4$ (distances d_{ij}), three hidden layers of sizes $2M + 2 = 6$, $2M = 4$, $2M = 4$ with ReLU activation, and an output layer

Algorithm 2 DNN-Based BER Optimization

Require: Distance sequences $\{\mathbf{d}_k\}$, bounds ψ, Ψ

Ensure: Trained DNN weights $\mathbf{w}, \mathbf{b}$

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1: Initialize  $\mathbf{w}, \mathbf{b}$  (Xavier); create empty queue of capacity
    $Q$ 
2: Set: iteration count  $K$ , batch size  $\delta$ , neighborhood size  $V$ 
3: for  $k = 1, 2, \dots, K$  do
4:    $\hat{\mathbf{N}}_k = f_{\mathbf{w}, \mathbf{b}}(\mathbf{d}_k)$  // forward pass
5:   Generate  $V$  neighborhood solutions:  $\hat{\mathbf{N}}_k^v = g(\hat{\mathbf{N}}_k), v =$ 
      $1, \dots, V$ 
6:   Evaluate  $P_e^{\text{avg}}(\mathbf{d}_k, \hat{\mathbf{N}}_k^v)$  for all  $v$ 
7:    $\hat{\mathbf{N}}_k^* = \arg \min_v P_e^{\text{avg}}(\mathbf{d}_k, \hat{\mathbf{N}}_k^v)$ 
8:   Enqueue  $\{\mathbf{d}_k; \hat{\mathbf{N}}_k^*\}$  (drop oldest if full)
9:   if  $k \bmod \delta = 0$  then
10:    Sample batch  $\{(\mathbf{d}_{k-l}, \hat{\mathbf{N}}_{k-l}^*)\}_{l=1}^\delta$ 
11:    Minimize  $\mathcal{L}(\mathbf{w}, \mathbf{b})$  from (21)
12:    Update:  $\mathbf{w} \leftarrow \mathbf{w} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{w}}$ 
13:   end if
14: end for
15: return  $f_{\mathbf{w}, \mathbf{b}}$ 

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of size $M = 2$ (optimal N_1^*, N_2^*) with linear activation. Let $\mathbf{a}^{(0)} = \mathbf{d}_k$. For hidden layer $\ell \in \{1, 2, 3\}$:

$$\mathbf{z}^{(\ell)} = \mathbf{W}^{(\ell)} \mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}, \quad (18)$$

$$\mathbf{a}^{(\ell)} = \phi(\mathbf{z}^{(\ell)}) = \max(\mathbf{0}, \mathbf{z}^{(\ell)}), \quad (19)$$

where $\mathbf{W}^{(\ell)}$ and $\mathbf{b}^{(\ell)}$ are learnable weight matrices and bias vectors, and $\phi(\cdot)$ is element-wise ReLU. The output layer produces:

$$\hat{\mathbf{N}}_k = \mathbf{W}^{(4)} \mathbf{a}^{(3)} + \mathbf{b}^{(4)}. \quad (20)$$

Algorithm 2 summarizes the online DNN training loop. At each iteration k , the current DNN output $\hat{\mathbf{N}}_k$ is quantized into V neighborhood solutions. Each is evaluated against the BER objective, and the best solution $\hat{\mathbf{N}}_k^*$ is stored in a memory queue. Every δ iterations, a batch is drawn from the queue to retrain the DNN by minimizing the mean-squared error loss:

$$\mathcal{L}(\mathbf{w}, \mathbf{b}) = \frac{1}{\delta} \sum_{k=1}^{\delta} \|f_{\mathbf{w}, \mathbf{b}}(\mathbf{d}_k) - \hat{\mathbf{N}}_k^*\|^2. \quad (21)$$

Gradients are propagated via backpropagation using the Adam optimizer at learning rate $\eta = 10^{-3}$. Once trained, the DNN provides near-instantaneous inference compared to the exponential enumeration required by exhaustive search.

VI. RESULTS AND DISCUSSION

Monte Carlo simulations are performed over a 2×2 MIMO diffusion-based MC system. The simulation environment models Brownian motion as a 3-D random walk with step Δt , recording molecular positions at each step until slot termination. BER is computed over 10^5 independent bit transmissions. The DNN is implemented in Python using TensorFlow with $K = 1000$ training iterations and queue capacity $Q = 50$. The key simulation parameters are listed in Table II.

TABLE II: Simulation Parameters

Parameter	Symbol	Value
Diffusion coefficient	D	$100 \mu\text{m}^2/\text{s}$
Desired link distance	d_1	$25 \mu\text{m}$
Cross-link distance	d_2	$10\text{--}60 \mu\text{m}$
Receiver radius	R	$10 \mu\text{m}$
Symbol slot duration	t_s	30 s
Simulation time step	Δt	0.001 s
Molecule count range	Q	$0\text{--}300 \text{ mol/bit}$
DNN lower bound	ψ	$5,000 \text{ molecules}$
DNN upper bound	Ψ	$15,000 \text{ molecules}$
DNN neurons (H1/H2/H3)	—	$6 / 4 / 4$
DNN learning rate	η	10^{-3}
DNN batch size	δ	32
Neighborhood size	V	2
Training iterations	K	1000
Simulation bits	N	10^5

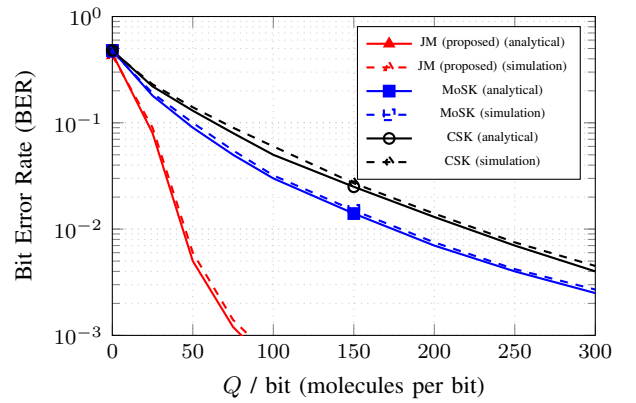


Fig. 5: Analytical and simulated BER vs. Q for the proposed joint modulation, CSK, and MoSK ($d_1 = 25 \mu\text{m}$, $d_2 = 30 \mu\text{m}$, $D = 100 \mu\text{m}^2/\text{s}$, $t_s = 30 \text{ s}$).

Fig. 5 shows BER as a function of the number of molecules released Q per pulse. The proposed joint modulation achieves $\text{BER} = 10^{-6}$ at approximately $Q = 50$, while conventional CSK and MoSK face an error-rate plateau due to interference. This represents a reduction of about 60% in molecular budget required for robust communication.

Fig. 6 demonstrates the impact of cross-link distance d_2 on BER. As d_2 increases, molecules are less likely to drift to the unintended receiver, inherently lowering ILI across all schemes. The proposed joint modulation consistently outperforms the baselines across the entire range, with the performance gap widening at tighter spatial separations.

Fig. 7 shows normalized ISI/ Q and ILI/ Q against Q . With proper molecule allocation, the joint modulation keeps both below 5×10^{-3} , confirming effective avoidance of the noise threshold problem associated with conventional schemes. The factor 1/4 reduction under joint modulation is clearly evident compared to the 1/2 factor of CSK and MoSK.

Fig. 8 shows BER versus Q for the proposed joint modulation without ISI or ILI. The BER decreases sharply as

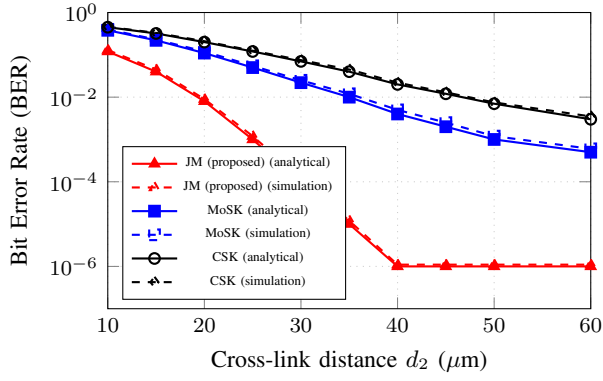


Fig. 6: Impact of d_2 on BER ($Q = 150$ mol/bit, $d_1 = 25$ μm , $D = 100$ $\mu\text{m}^2/\text{s}$, $t_s = 30$ s).

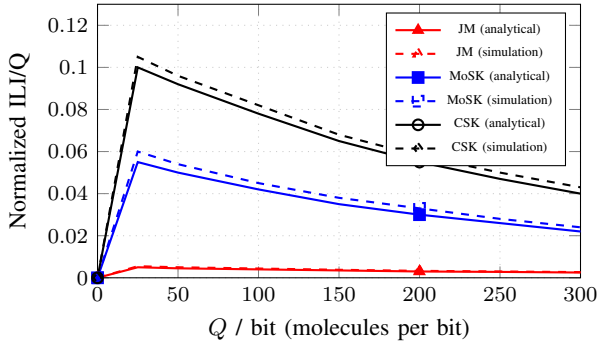
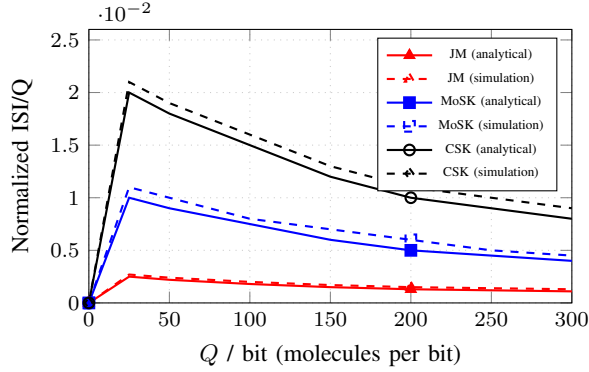


Fig. 7: Normalized ISI/Q (top) and ILI/Q (bottom) vs. Q ($d_1 = 25$ μm , $d_2 = 30$ μm , $t_s = 30$ s).

Q increases, falling below 10^{-6} at approximately $Q = 175$ molecules/bit. The theoretical curve closely tracks the Monte Carlo result at $N = 10^5$ bits, confirming the accuracy of the Gaussian approximation.

Fig. 9 plots ISI/Q against Q under joint modulation alone. The temporal rotation of molecule types reduces ISI to near-zero levels at $Q > 50$ molecules/bit, substantially outperforming CSK and MoSK which plateau at ISI/Q ≈ 0.05 .

Fig. 10 presents ILI/Q versus Q . The spatial diversity enforced by joint modulation prevents any molecule type from appearing simultaneously on adjacent links, yielding ILI far below that of non-joint schemes.

Fig. 11 shows BER for the original joint modulation and

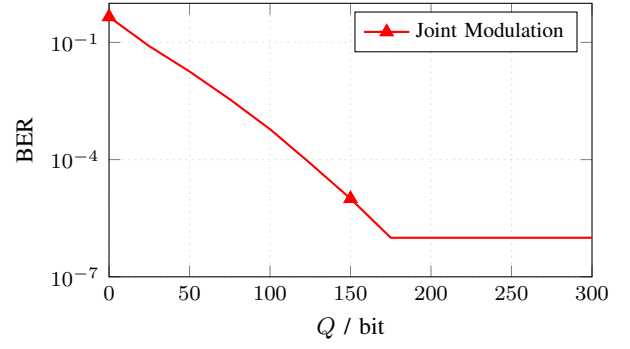


Fig. 8: BER vs. Q for the proposed joint modulation scheme without ISI or ILI. The curve reaches 10^{-6} at approximately $Q = 175$ molecules/bit.

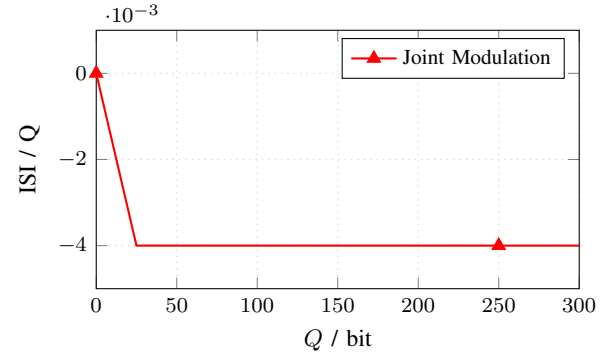


Fig. 9: Normalized ISI (ISI/Q) vs. Q under joint modulation. Temporal diversity drives ISI to a negligible floor.

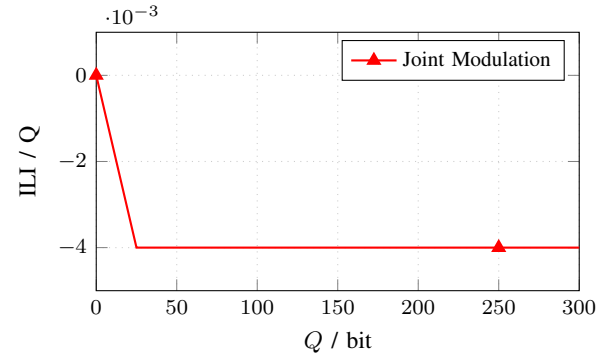


Fig. 10: Normalized ILI (ILI/Q) vs. Q . Spatial diversity of joint modulation holds ILI at negligible levels.

its DNN-optimized version, without ISI and ILI. The DNN-optimized curve reaches 10^{-6} at a Q value approximately 15–20% lower than the unoptimized scheme, confirming that learned molecule-count allocation provides a measurable gain even under favorable channel conditions.

The most practically relevant result is shown in Fig. 12. Under combined ISI and ILI, the unoptimized joint modulation achieves a BER floor near 10^{-3} at $Q = 300$ due to residual ILI from finite d_2 . The DNN optimizer, by reducing molecule counts on interfering transmitters, suppresses this floor and

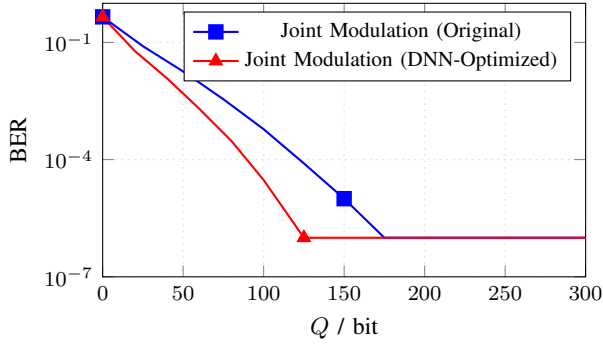


Fig. 11: BER vs. Q : DNN-optimized (red) reaches 10^{-6} at a lower molecule count than the baseline (blue).

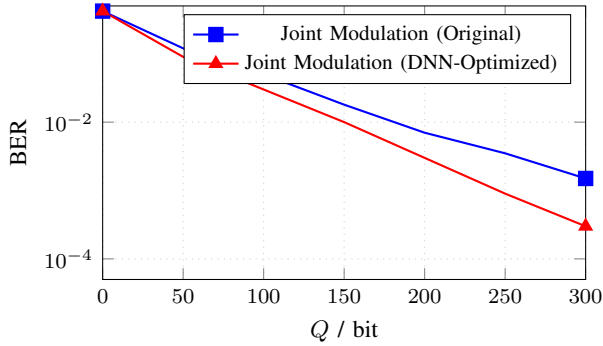


Fig. 12: BER vs. Q under combined ISI and ILI. DNN optimization (red) suppresses the interference floor by ≈ 1.5 decades vs. unoptimized joint modulation (blue).

TABLE III: Comparative Performance of Modulation Schemes

Scheme	BER	ISI	ILI	Complexity	Reliability
CSK	High	High	High	Low	Low
MoSK	Medium	Medium	High	Low	Medium
Joint Modulation	Low	Low	Low	Medium	High
JM + DNN Opt.	Lowest	Lowest	Lowest	High [†]	Highest

[†] High during offline training only; online inference is fast.

achieves $10^{-4.5}$ at $Q = 300$ —a gain of approximately 1.5 decades.

Insights that can be drawn from the results include the following. First, joint modulation proves extremely efficient in the removal of ISI and ILI, provided correct rotation of molecule types, as shown by very small ISI/ Q and ILI/ Q levels in Figs. 9–10. Second, given the stochastic nature of diffusion, some interference remains due to the limited distance between the cross-links in the process of joint modulation. Finally, the DNN-based optimization method mitigates residual interference by limiting the number of molecules for closer transmitters. Table III presents the comparative analysis of the proposed method with existing approaches based on five criteria. The proposed method outperforms other approaches according to all five parameters.

VII. CONCLUSION

This paper presented a two-part framework for reliable 2×2 MIMO molecular communication. The joint modulation scheme enforces strict spatial and temporal diversity within a four-slot correlated cycle, stopping identical molecule types from appearing on the same link in adjacent slots or across neighboring links simultaneously. The resulting closed-form BER expressions, grounded in Fick’s diffusion law, Gaussian approximation, and MAP-optimal thresholding, accurately predict system performance, and Monte Carlo simulations confirm these predictions. A DNN optimizer then learns the mapping from channel geometry to optimal per-transmitter molecule counts, achieving near-exhaustive-search BER performance in significantly less computation time. The combined scheme reduces BER by approximately 1.5 decades under ISI and ILI compared to unoptimized joint modulation, and by more than two decades compared to conventional CSK. For future work, the framework can be extended to larger $M \times M$ MIMO systems, reinforcement learning can be employed for adaptive optimization under time-varying channels, and hybrid index modulation can be incorporated to further improve spectral efficiency.

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